Quantitative Detection Limits for Cosmic Muon Scattering Tomography in Infrastructure Inspection: An Analytical Feasibility Framework

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Abstract

We present an analytical framework for estimating detection time requirements in cosmic muon scattering tomography applied to reinforced concrete infrastructure inspection. Using the Highland multiple Coulomb scattering formalism with correct per-track signal calculation, realistic detector angular resolution (3 mrad/track), and structural noise from baseline density variation, we compute detection significance as a function of exposure time for three defect classes in a 2 m thick concrete target: planar air cracks (5 mm), spherical voids (10 mm), and rebar surface corrosion (2 mm rust layer).

Results: (1) A 5 mm full-width planar crack reaches 3-sigma detection at approximately 23 days and 5-sigma at approximately 67 days of continuous exposure; (2) a 10 mm spherical void requires approximately 197 days for 3-sigma due to its small geometric cross-section; (3) rebar corrosion (2 mm) is impractical at sea level (>1 year). These timescales are consistent with published Geant4 simulations reporting month-long exposures for concrete defect imaging [Tripathy et al. 2021] and 30-day exposures for corrosion detection [Yilmaz et al. 2022]. The critical finding is that planar defects (cracks, delaminations) are far more detectable than point defects (voids) of similar dimension because planar geometry intercepts a larger fraction of muon tracks. We identify the dominant noise source as detector angular resolution (3 mrad), not Poisson counting statistics, and show that detection time scales as t proportional to 1/(delta-theta-squared times f_geo), where delta-theta is the per-track scattering change and f_geo is the geometric track fraction. The framework provides a simple, reproducible order-of-magnitude feasibility assessment tool for infrastructure stakeholders evaluating muon tomography deployment.

Prior analytical frameworks for muon imaging feasibility (Lesparre et al. 2010; Kodama and Matsushima 2021) address transmission (absorption) muography. To our knowledge, this work provides the first analogous closed-form framework for scattering tomography applied to concrete infrastructure defects.

Keywords: muon scattering tomography, structural health monitoring, non-destructive testing, cosmic rays, infrastructure inspection, detection limits, multiple Coulomb scattering

1. Introduction

1.1 The Infrastructure Problem

Aging civil infrastructure represents a global engineering challenge. In the United States, the American Society of Civil Engineers (ASCE) rates 42% of bridges as fair or worse, with over 45,000 classified as structurally deficient [1]. In Europe, over one million concrete bridges exceed 50 years of age, approaching or exceeding their design lifetime [2]. Dam safety presents comparable concerns: the U.S. National Inventory of Dams lists over 9,000 facilities, many constructed before modern safety standards [3].

Current non-destructive evaluation (NDE) methods are fundamentally depth-limited. Ultrasonic testing penetrates approximately 40 cm in reinforced concrete and suffers from rebar reflections [4, 5]. Ground-penetrating radar (GPR) achieves similar depths with signal attenuation in high-moisture concrete [6]. Visual inspection and tap testing detect only surface distress [7]. No cost-effective method provides continuous, deep (>1 m) assessment of internal structural condition.

1.2 Muon Tomography: Physical Basis

Cosmic ray muons are secondary particles produced by primary cosmic ray interactions in the upper atmosphere [8]. At sea level, the vertical muon flux is approximately 1 muon/(cm-squared per minute) with a mean momentum of approximately 4 GeV/c [9, 10]. Muons penetrate several meters of concrete and tens of meters of rock before absorption [11].

When traversing matter, muons undergo multiple Coulomb scattering -- deflection by atomic nuclei. The RMS scattering angle is described by the Highland formula [12, 13]:

```
theta_rms = (13.6 MeV / pc) * sqrt(L / X0) * (1 + 0.038 * ln(L/X0))
[Eq. 1]
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where p is the muon momentum, L is the material thickness, and X0 is the radiation length. Published radiation lengths [14]: concrete X0 = 26.7 cm, steel (iron) X0 = 1.76 cm, air X0 = 3.04 x 10^4 cm. Denser materials scatter muons more strongly; a void (air) produces negligible scattering, creating a detectable contrast.

1.3 Prior Art

1.3.1 Foundational Work

Muon scattering tomography was proposed by Borozdin et al. at Los Alamos National Laboratory in 2003 [15]. This foundational work demonstrated that the scattering angle distribution of cosmic muons passing through a target encodes material composition, enabling non-invasive imaging of dense objects. The Point of Closest Approach (PoCA) algorithm for reconstruction was introduced concurrently [16].

1.3.2 Geological and Archaeological Applications

Muon radiography (absorption-based) has been applied to volcanoes [17, 18], the Khufu pyramid [19], and underground voids [20]. These applications demonstrate penetration capabilities of tens to hundreds of meters in geological media, establishing the physical viability of muon-based imaging at infrastructure scales.

1.3.3 Analytical Feasibility Frameworks

Lesparre et al. (2010) derived an analytical formula for the relationship between exposure time and density resolution for muon transmission (absorption) radiography [18]. Their framework quantifies the minimum observation time required to distinguish different density-length contrasts under Gaussian statistics. Kodama and Matsushima (2021) extended this to a three-way trade-off among spatial resolution, temporal resolution, and density resolution [41]. Kodama et al. (2023) further investigated anomaly detectability in multidimensional muon tomography using inversion techniques and machine learning [42]. These frameworks address transmission muography, where the observable is muon flux attenuation. The present work provides the first analogous framework for scattering tomography, where the observable is the angular deflection distribution.

1.3.4 Infrastructure Applications

Niederleithinger et al. (2021) published the first experimental proof-of-concept for muon tomography of reinforced concrete blocks [21]. They demonstrated detection of individual reinforcement bars (33.7 +/- 7.3 mm diameter) at 50 cm depth. Tripathy et al. (2021) performed Geant4 simulations of muon scattering tomography for concrete defect imaging, using 30-day cosmic muon exposure [22]. Yilmaz et al. (2022) applied machine learning to muon tomography for rebar corrosion identification, reporting that 30% rust could be identified with 30-day exposure and 7.3% error [23]. Cimmino et al. (2025) combined deep learning with muon tomography for structural diagnostics [24].

1.3.5 Commercial Deployment

GScan (Estonia), founded by CERN physicists in 2018, has deployed muon scanners on bridges in Estonia, the UK, and the Middle East [25, 26]. Muon Vision (Finland) applies muon tomography to tailings dam monitoring and underground mining [29]. MuRayTech, a GScan spinout, is developing laser-plasma accelerator technology for artificial muon generation [31], which would bypass the cosmic flux limitation entirely.

1.4 Gap Addressed by This Work

Prior studies provide individual detection measurements for specific scenarios using computationally intensive Geant4 simulations. Existing analytical frameworks (Lesparre et al. 2010; Kodama and Matsushima 2021) address transmission muography, not scattering tomography. What is lacking is: (1) A simple analytical framework that predicts scattering-based detection time from first principles without requiring Geant4; (2) Systematic comparison of multiple defect types under identical conditions; (3) Explicit identification of the dominant noise source (detector resolution vs. Poisson statistics); (4) A practical order-of-magnitude feasibility assessment tool for infrastructure stakeholders evaluating muon scattering tomography deployment. This work provides all four.

2. Methods

2.1 Detector Model

We model a portable muon tomography system based on published GScan specifications [25] and the BAM experimental configuration [34]: Detector area 1.2 m-squared (1.0 m x 1.7 m), two planes (above and below target), detector efficiency 70%, angular acceptance +/-30 degrees halfangle, angular resolution 3 mrad per track (200 micron position resolution at 7 cm gap [22]), target thickness 200 cm reinforced concrete, target cross-section 100 x 100 cm.

2.2 Track Accumulation Rate

```
R = Phi x A x epsilon x T   [Eq. 2]
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where Phi = 1 muon/(cm-squared per minute), A = 12,000 cm-squared, epsilon = 0.70, and T = exp(-200/180) = 0.329 (transmission through 200 cm concrete with interaction length 180 cm [36]). This gives R = 2,765 tracks/min, approximately 4.0 x 10^6 tracks/day. We note that the transmission factor uses nuclear interaction length; accounting for the full energy spectrum and electromagnetic stopping would yield T approximately 0.5-0.6, making our track rate estimate conservative.

2.3 Signal Model

The signal is the per-track scattering angle change for muons traversing the defect region compared to bulk material.

2.3.1 Per-Track Scattering Change

A muon passing through bulk concrete (200 cm) scatters with RMS angle:

```
theta_bulk = (0.0136 / 4.0) * sqrt(200 / 26.7) * (1 + 0.038 *
ln(200/26.7)) = 10.02 mrad   [Eq. 3]
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A muon passing through the same column but with a 5 mm (0.5 cm) air crack at mid-depth traverses 199.5 cm of concrete and 0.5 cm of air. Scattering adds in quadrature (independent processes):

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theta_with_crack = sqrt(theta_concrete_199.5^2 + theta_air_0.5^2) =
sqrt(10.00^2 + 0.003^2) = 10.00 mrad   [Eq. 4]
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Per-track signal: delta-theta = |theta_bulk - theta_with_crack| = 0.013 mrad [Eq. 5]. This is 0.13% of the bulk scattering. The small fractional change is why detection requires many tracks and extended exposure.

2.3.2 Geometric Track Fraction

The fraction of detected muons that pass through the defect region depends on defect geometry:

Planar defect (crack): A horizontal crack of width w extending across the full column width intercepts all tracks passing through its plane: f_planar = w / (W * cos(theta)), where W is the target width. For w = 0.5 cm, W = 100 cm: f_planar = 0.0052 (0.52%). We note this assumes the crack extends across the full cross-section. Partial-width cracks would have proportionally smaller geometric fractions and correspondingly longer detection times.

Point defect (void): A sphere of diameter d_v intercepts tracks proportional to its projected area: f_point = pi * (d_v/2)^2 / W^2. For d_v = 1.0 cm: f_point = 7.85 x 10^-5 (0.008%).

Key insight: Planar defects have geometric fractions approximately 66 times larger than point defects of the same dimension. This explains why cracks are far more detectable than isolated voids.

2.4 Noise Model

For N_ROI tracks passing through the defect region, the uncertainty in their mean scattering angle is sigma_mean = sigma_eff / sqrt(N_ROI), where the effective per-track noise combines detector resolution and structural noise:

```
sigma_eff = sqrt(sigma_detector^2 + sigma_structural^2)   [Eq. 9]
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sigma_detector = 3.0 mrad (typical scintillator hodoscope [22, 35]); sigma_structural = 0.05 x theta_bulk = 0.50 mrad (5% baseline density variation [37]). This gives sigma_eff = sqrt(3.0^2 + 0.50^2) = 3.04 mrad. Note: sigma_eff is approximately equal to sigma_detector. Detector angular resolution dominates.

Improvement pathway: Reducing angular resolution from 3 mrad to 1 mrad would reduce detection times by a factor of approximately 7.4 (since t is proportional to sigma_eff^2, and at 1 mrad detector resolution, sigma_eff = sqrt(1.0^2 + 0.50^2) = 1.12 mrad, giving (3.04/1.12)^2 = 7.4). This is the most impactful engineering improvement.

2.5 Detection Significance Formula

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sigma_detection = delta_theta * sqrt(f_geo * N_total) / sigma_eff
[Eq. 10]
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For target significance sigma_0 (e.g., 5-sigma):

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N_total = (sigma_0 * sigma_eff)^2 / (delta_theta^2 * f_geo)   [Eq. 11]
t_exposure = N_total / R   [Eq. 12]
```

This is the central result of the paper. All detection times follow from Equations 10-12. We note that this model computes statistical detection significance (distinguishing defect-region mean scattering from bulk mean scattering), not image reconstruction quality. The model assumes a binary comparison and does not exploit spatial correlation across pixels, making it conservative relative to full tomographic reconstruction.

2.6 Rebar Proximity Caveat

Real reinforced concrete contains steel rebar grids. Steel (X0 = 1.76 cm) scatters muons approximately 15 times more per unit length than concrete (X0 = 26.7 cm). A muon passing through a 1 cm rebar produces comparable scattering to passing through 15 cm of concrete. If a defect is located adjacent to or between rebar elements, the local scattering background differs from the bulk concrete assumption. This effect can either help detection (rebar creates additional contrast near voids) or hinder it (rebar noise masks small defect signals). The present framework does not model rebar-defect interaction and assumes a uniform concrete matrix. Rebar effects should be evaluated with Geant4 simulation for specific inspection geometries.

3. Results

3.1 Defect Parameters

Table 1. Defect parameters and detection characteristics.

Defect	Thickness	X0	d-theta (mrad)	f_geo	Contrast
5 mm air crack	0.5 cm	30,400 cm	0.0134	0.00524	0.13%
10 mm void	1.0 cm	30,400 cm	0.0268	0.000157	0.27%
Rebar corrosion (2mm)	0.2 cm	2.88 cm	0.0270	0.000145	0.27%

3.2 Detection Times

Table 2. Exposure time to achieve target significance.

Defect	3-sigma	5-sigma	7-sigma
5 mm air crack (full-width)	23 days	67 days	124 days
10 mm void	197 days	>1 year	>1 year
Rebar corrosion (2mm)	230 days	>1 year	>1 year

3.3 Validation Against Published Literature

Table 3. Comparison with published results.

Published Result	Our Prediction	Assessment
Tripathy et al. [22]: 30-day exposure, Geant4	23 days for 3-sigma crack	Consistent (similar timescale)

Yilmaz et al. [23]: 30 days, 30% corrosion, ML	>1 year for 2 mm corrosion	Consistent (2 mm far subtler than 30%)
Georgadze [32]: 16.4 days, 3-sigma, CORSIKA+Geant4	23 days for 3-sigma crack	Consistent (our model more conservative)

3.4 Defect Size Scaling

Using Equations 10-12, we compute required exposure for 5-sigma detection as a function of crack width: 1 mm crack approximately 800 days (impractical); 5 mm crack approximately 67 days (feasible for permanent installation); 10 mm crack approximately 17 days (practical for periodic scans); 20 mm crack approximately 4 days (practical for rapid assessment); 50 mm delamination less than 1 day (comparable to GScan reported timescales). This scaling law explains why GScan can detect large post-tensioning duct features (>20 mm) in hours-to-days, while our model predicts weeks for 5 mm cracks. Both are correct for their respective defect sizes.

4. Discussion

4.1 Why Cracks Are More Detectable Than Voids

The counterintuitive result -- a 5 mm crack is detected 3-10 times faster than a 10 mm void -- follows directly from geometric track fraction. A crack with planar geometry has f proportional to w/W, so many tracks cross it. A void with point geometry has f proportional to d-squared/W-squared, so very few tracks cross it. A horizontal crack extends across the entire column; every muon passing within +/-2.5 mm of the crack plane is affected. A 10 mm sphere presents a tiny target (approximately 0.008% of the cross-section).

Infrastructure implication: Muon scattering tomography is naturally well-suited for detecting delaminations, disbonds, and planar features, which are among the most dangerous failure modes in post-tensioned concrete (tendon duct voids, grout failures, interface delamination [38]).

4.2 The Detector Resolution Bottleneck

The dominant noise source is detector angular resolution (3 mrad), not counting statistics. The signal per track (delta-theta = 0.013 mrad for a 5 mm crack) is 230 times smaller than the measurement noise. Detection proceeds purely by statistical averaging over millions of tracks.

Reducing angular resolution from 3 mrad to 1 mrad would reduce detection times by a factor of approximately 7.4 (accounting for the structural noise floor: at 1 mrad detector resolution, $\sigma_{eff} = \sqrt{1.0^2 + 0.50^2} = 1.12$ mrad, giving $(3.04/1.12)^2 = 7.4$). High-resolution drift chambers achieve approximately 350 micron position resolution [33], corresponding to approximately 1 mrad angular resolution at typical detector spacing. This engineering improvement would shift 5 mm crack detection from approximately 67 days to approximately 9 days, consistent with GScan's reported multi-day scan durations for comparable defect sizes.

4.3 Corrosion: A Definitive Negative Result

Rebar corrosion detection (2 mm rust layer) is impractical at sea level. The combined effect of tiny geometric fraction (0.015% of cross-section), small scattering contrast (rust X_0 = 2.88 cm vs. concrete X_0 = 26.7 cm), and detector noise domination produces detection times exceeding one year. Yilmaz et al. [23] required 30 days to detect 30% corrosion -- massive degradation representing advanced structural failure. Early-stage corrosion is far subtler. This negative result should guide infrastructure owners: do not procure muon tomography primarily for corrosion detection. Alternative methods (acoustic emission sensors [39], electrochemical half-cell measurements [40]) detect corrosion at earlier stages.

4.4 Rebar Interaction Effects

The present framework assumes a uniform concrete matrix and does not model the interaction between defects and nearby steel reinforcement. In real structures, rebar grids create local scattering hot spots that complicate defect detection. A muon passing through 1 cm of steel (X_0 = 1.76 cm) scatters comparably to passing through approximately 15 cm of concrete. This has two consequences: (1) voids adjacent to rebar may be easier to detect because the steel-to-void contrast is larger than concrete-to-void contrast; (2) in heavily reinforced regions, the spatial variation of baseline scattering (from rebar at different positions) increases the effective structural noise beyond the 5% assumed here. Geant4 simulation of specific rebar geometries is recommended for operational planning.

4.5 The Baseline Subtraction Advantage

Our model assumes absolute detection -- identifying a defect against a uniform density background. In practice, change detection (comparing two scans over time) is more powerful because baseline subtraction removes structural noise. With baseline subtraction, sigma_eff reduces from 3.04 mrad to approximately 3.0 mrad (detector resolution only) -- a modest improvement for current detectors. However, for future high-resolution detectors (sigma_detector < 1 mrad), structural noise would become dominant, and baseline subtraction would provide a substantial advantage. This is the optimal deployment model: install permanently, establish baseline over 4-8 weeks, detect changes over months and years.

4.6 Comparison with Existing NDE Methods

Muon tomography does not replace ultrasound or GPR. It fills the depth gap that neither can reach. Ultrasound penetrates approximately 40 cm with 2-4 hour scan times and is best for shallow defects. GPR reaches approximately 50 cm with similar scan times and excels at rebar location. Muon scattering tomography penetrates 10+ meters but requires weeks for 5 mm defect detection at sea level. The optimal inspection system combines shallow-fast methods (ultrasound, GPR) with deep-slow methods (muons).

4.7 Relationship to Transmission Muography Frameworks

Lesparre et al. (2010) [18] derived analytical detection limits for muon transmission radiography, relating exposure time to the minimum resolvable density-length contrast. Kodama and Matsushima (2021) [41] extended this to include anomaly size as a third parameter. These frameworks address a fundamentally different observable: muon flux attenuation (counting how many muons pass through). The present work addresses muon angular scattering (measuring how much muons deflect). The key physical differences are: (1) scattering tomography uses both incoming and outgoing track angles, enabling 3D localization via PoCA or MLEM reconstruction; (2) scattering signal scales with radiation length X0 rather than density alone, providing material-type discrimination; (3) the dominant noise source in scattering tomography is detector angular resolution, whereas in transmission muography it is Poisson counting statistics. Despite these physical differences, the mathematical structure is analogous: detection significance scales as signal times square root of exposure time, divided by per-measurement noise.

4.8 Limitations

(1) Analytical model vs. Monte Carlo: We use a closed-form model rather than Geant4. This is intentional but means we do not capture second-order effects (energy spectrum, multiple scattering correlations, detector edge effects). Results are order-of-magnitude feasibility estimates, not precision predictions.

(2) Idealized defect geometry: Real cracks are irregular, voids have rough boundaries, and corrosion is distributed. Our planar crack model assumes a full-width crack. Partial-width cracks would require proportionally longer detection times. Real-world detection may require longer exposure than predicted here.

(3) Single muon energy: We use the spectrum-averaged momentum of 4 GeV/c. A proper treatment would integrate over the sea-level energy spectrum. Low-energy muons (1-3 GeV) scatter more and still penetrate 2 m of concrete, so the spectrum-averaged signal is likely larger than our single-energy estimate, making our detection times conservative. The effect is estimated at 20-40% [22].

(4) No reconstruction algorithm modeled: We compute statistical detection significance, not image quality. Reconstruction algorithms (PoCA, MLEM) introduce their own noise but also exploit spatial correlation, and these effects roughly offset.

(5) Uniform matrix assumption: The model does not account for rebar grids, aggregate variation, or moisture gradients. See Section 4.4 for discussion of rebar effects.

5. Conclusions

5.1 Principal Findings

(1) A 5 mm full-width planar crack in 2 m concrete is detectable at 3-sigma in approximately 23 days of continuous cosmic muon exposure at sea level, using a 1.2 m-squared detector with 3 mrad angular resolution. This is consistent with published Geant4 simulations [22, 32].

(2) Point defects (10 mm voids) require months due to their small geometric cross-section (approximately 0.008%), despite producing larger per-track scattering changes than cracks. Planar defects are inherently more detectable.

(3) Rebar corrosion (2 mm surface rust) is impractical at sea level. This definitive negative result should discourage procurement of muon tomography systems primarily for corrosion monitoring.

(4) The dominant noise source is detector angular resolution (3 mrad), not Poisson statistics. Improving resolution to 1 mrad would reduce all detection times by approximately 7.4 times (accounting for the structural noise floor), making weekly crack detection feasible.

(5) Passive long-term monitoring is the correct deployment model. Install permanently, establish baseline over 4-8 weeks, detect changes over months and years. Muon scattering tomography fills the depth gap (>1 m) that ultrasound and GPR cannot reach.

5.2 The Detection Time Formula

The central deliverable of this paper is a simple, closed-form detection time estimate:

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t_exposure = sigma_0^2 * sigma_eff^2 / (delta_theta^2 * f_geo * R)
[Eq. 13]
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where sigma_0 is the target significance (e.g., 5), sigma_eff is approximately 3 mrad (detector resolution), delta_theta is the per-track scattering change (Equation 5), f_geo is the geometric track fraction (Section 2.3.2), and R is the track rate (Equation 2). This formula allows any infrastructure engineer to obtain an order-of-magnitude feasibility estimate for their specific structure, defect type, and detector configuration without requiring Geant4 expertise. We emphasize that results from this framework are analytical estimates suitable for feasibility assessment, not substitutes for detailed Monte Carlo simulation in operational planning.

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